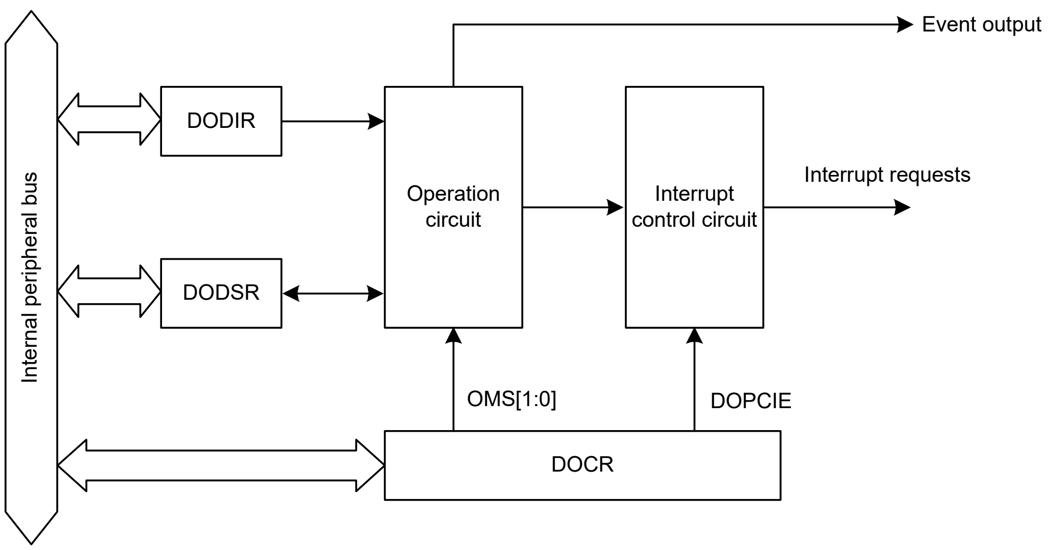
59. Data Operation Circuit (DOC)

59.1 Overview

The data operation circuit (DOC) is used to compare, add, or subtract 16-bit values.
Table 59.1 lists the specifications of the DOC and Figure 59.1 is a block diagram of the DOC.
An interrupt can be generated if the result of 16-bit comparison meets one of the set interrupt conditions.

Table 59.1 DOC Specifications

Item	Description
Data operation function	16-bit data comparison, addition, and subtraction
Lower power consumption function	The DOC can be placed in a module-stop state.
Interrupts	• The result of data comparison meets the detection condition. • The result of data addition is greater than FFFFh, which is an overflow. • The result of data subtraction is less than 0000h, which is an underflow.
Event link function (output)	• The result of data comparison meets the detection condition. • The result of data addition is greater than FFFFh, which is an overflow. • The result of data subtraction is less than 0000h, which is an underflow.



DOCR: DOC control register
DODIR: DOC data input register
DODSR: DOC data setting register

Figure 59.1 DOC Block Diagram

59.2 Register Descriptions

59.2.1 DOC Control Register (DOCR)

Address(es): DOC.DOCR 0008 B080h

	b7	b6	b5	b4	b3	b2	b1	b0
	—	DOPCFCL	DOPCF	DOPCIE	—	DCSEL	OMS[1:0]	
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b1, b0	OMS[1:0]	Operating Mode Select	b1 b0 0 0: Data comparison mode 0 1: Data addition mode 1 0: Data subtraction mode 1 1: Setting prohibited	R/W
b2	DCSEL	Detection Condition Select*1	0: 'Not equal to' is to be detected. 1: 'Equal to' is to be detected.	R/W
b3	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b4	DOPCIE	Data Operation Circuit Interrupt Enable	0: Interrupt disabled 1: Interrupt enabled	R/W
b5	DOPCF	Data Operation Result Flag	Indicates the result of an operation.	R
b6	DOPCFCL	Data Operation Result Clear	0: Retain the value of the DOPCF flag. 1: Clears the DOPCF flag.	R/W
b7	—	Reserved	This bit is read as 0. The write value should be 0.	R/W

Note 1. Valid only when data comparison mode is selected.

The DOCR register specifies the operation of DOC, or enabling or disabling of the interrupt.

OMS[1:0] Bits (Operating Mode Select)

These bits select the operating mode of the DOC.

DCSEL Bit (Detection Condition Select)

This bit is valid only when data comparison mode is selected.
This bit selects the condition for detection in data comparison mode.

DOPCIE Bit (Data Operation Circuit Interrupt Enable)

Setting this bit to 1 enables interrupts from the DOC.

DOPCF Flag (Data Operation Result Flag)

[Setting conditions]

- The condition selected by the DCSEL bit is met
- A result of data addition is greater than FFFFh
- A result of data subtraction is less than 0000h

[Clearing condition]

- Writing 1 to the DOPCFCL bit

DOPCFCL Bit (Data Operation Result Clear)

Writing 1 to this bit clears the DOPCF flag.

This bit is read as 0.

59.2.2 DOC Data Input Register (DODIR)

Address(es): DOC.DODIR 0008 B082h

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

The DODIR register is a readable and writable register that holds values for use in operations.

59.2.3 DOC Data Setting Register (DODSR)

Address(es): DOC.DODSR 0008 B084h

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

The DODSR register is a readable and writable register that holds values for use in comparison or the results of other operations.

In data comparison mode, store the standard value for use in comparison in this register.

In data addition or data subtraction mode, this register holds the results of operations.

59.3 Operation

59.3.1 Data Comparison Mode

Figure 59.2 shows an example of the steps involved in data comparison mode operation by the DOC.
An example of operation when DCSEL is set to 0 (‘not equal to’ is to be detected as the result of data comparison) is shown below.

(1) Writing 00b to the DOCR.OMS[1:0] bits places the DOC in the data comparison mode.
(2) Specify the standard value for comparison in the DODSR register.
(3) Write the value for comparison in the DODIR register.
(4) Write all values for use in comparison to the DODIR register.
(5) If the value written to the DODIR register is not equal to the value set in the DODSR register, the DOCR.DOPCF flag becomes 1. If the DOCR.DOPCIE bit is 1, a data operation circuit interrupt is also issued.

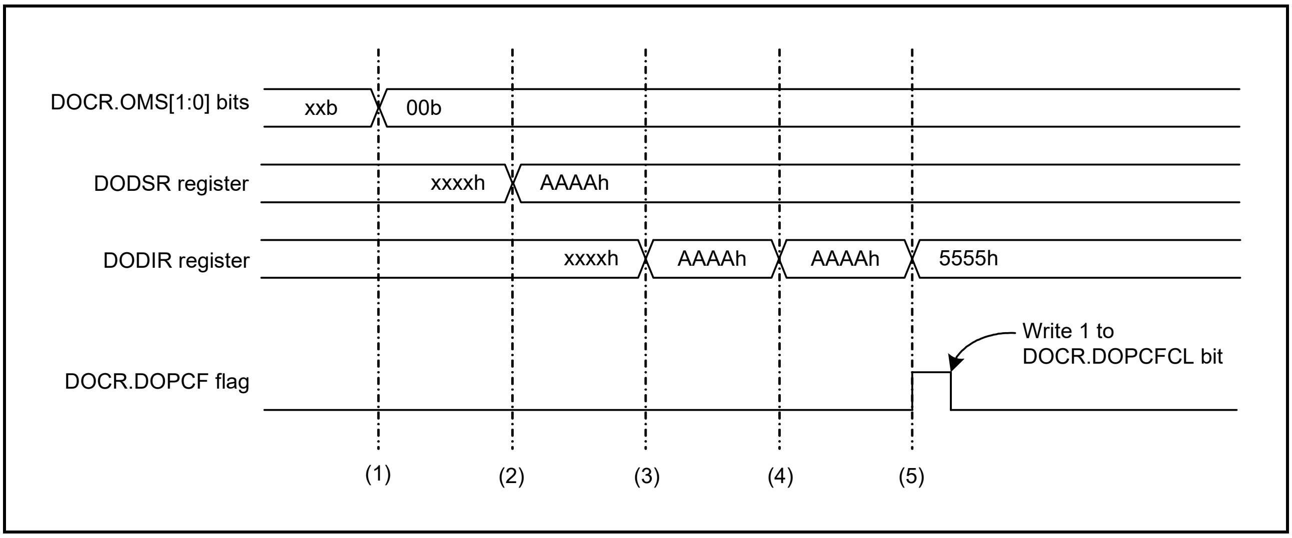


Figure 59.2 Example of Operation in Data Comparison Mode

59.3.2 Data Addition Mode

Figure 59.3 shows an example of the steps involved in data addition mode operation by the DOC.

(1) Writing 01b to the DOCR.OMS[1:0] bits selects data addition mode.
(2) Set the initial value in the DODSR register.
(3) Write the value for addition in the DODIR register. The result of the operation is stored in DODSR.
(4) Write all values for use in addition to the DODIR register.
(5) If the result of the operation is greater than FFFFh, the DOCR.DOPCF flag becomes 1. If the DOCR.DOPCIE bit is 1, a data operation circuit interrupt is also issued.

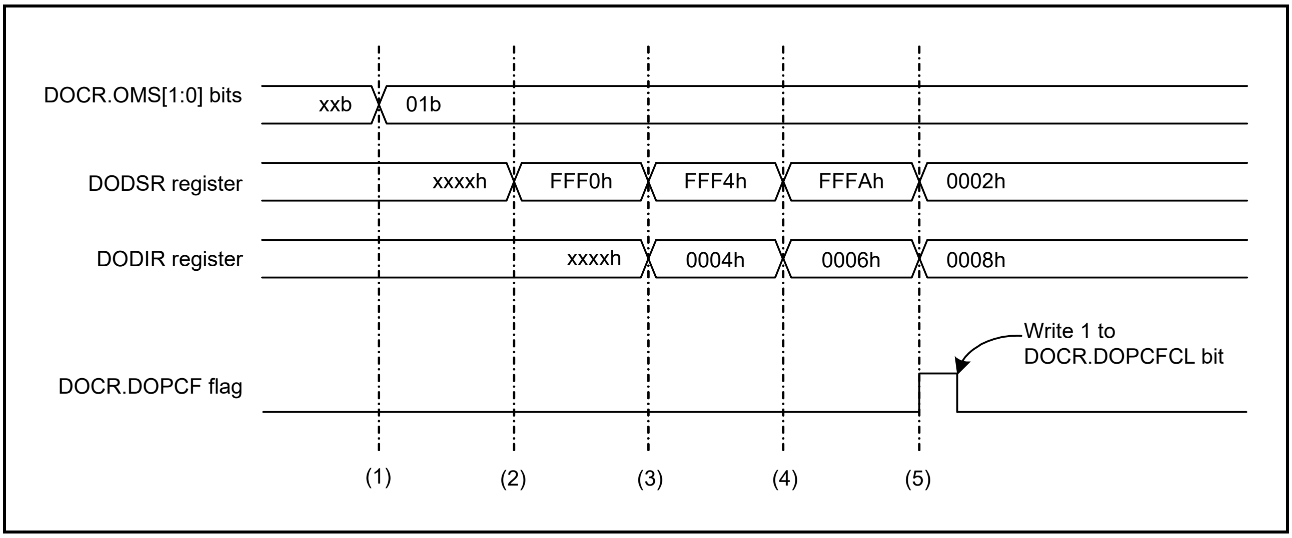


Figure 59.3 Example of Operation in Data Addition Mode

59.3.3 Data Subtraction Mode

Figure 59.4 shows an example of the steps involved in data subtraction mode operation by the DOC.

(1) Writing 10b to the DOCR.OMS[1:0] bits selects data subtraction mode.
(2) Set the initial value in the DODSR register.
(3) Write the value for subtraction in the DODIR register. The result of the operation is stored in DODSR.
(4) Write all values for use in subtraction to the DODIR register.
(5) If the result of the operation is less than 0000h, the DOCR.DOPCF flag becomes 1. If the DOCR.DOPCIE bit is 1, a data operation circuit interrupt is also issued.

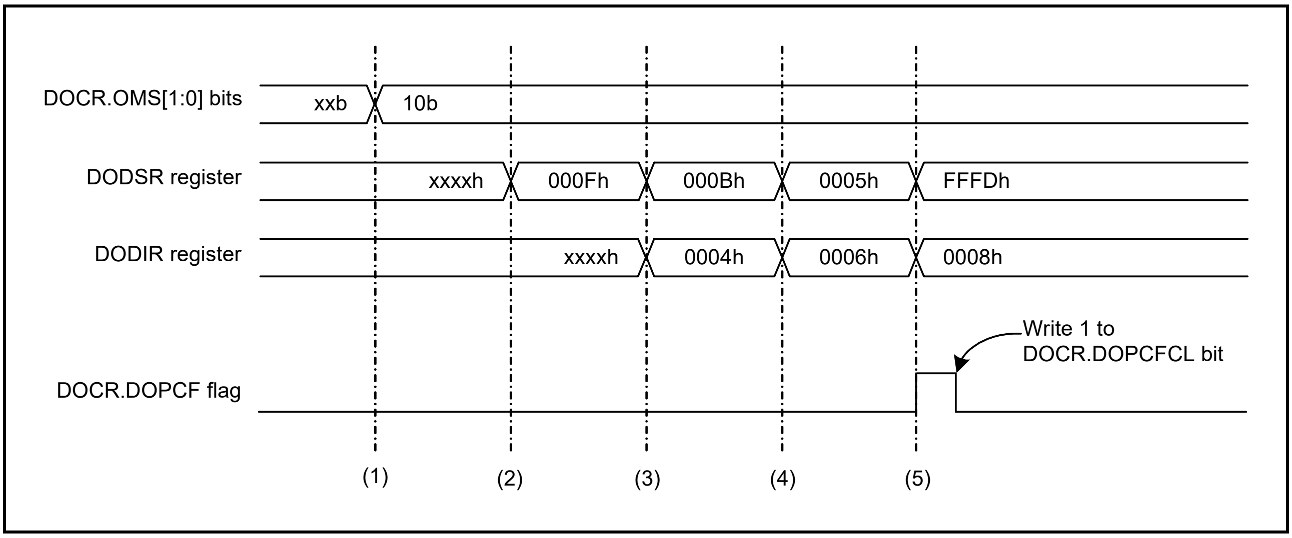


Figure 59.4 Example of Operation in Data Subtraction Mode

59.4 Interrupt Requests

The data operation circuit interrupt (DOPCI) is the interrupt request generated by the DOC. The DOCR.DOPCF flag becomes 1 when the interrupt source condition is satisfied.
Table 59.2 lists the details of the interrupt request.

Table 59.2 Interrupt Request from DOC

Interrupt Request	Data Operation Result Flag	Interrupt Generation Timing
Data operation circuit interrupt (DOPCI)	DOPCF	• The result of data comparison meets the detection condition. • The result of data addition is greater than FFFFh. • The result of data subtraction is less than 0000h.

59.5 Event Link Output

The DOC outputs event signals for the event link controller (ELC) under the following conditions, and these can be used to initiate operations by other modules selected in advance.

- The result of data comparison meets the detection condition.
- The result of data addition is greater than FFFFh.
- The result of data subtraction is less than 0000h.

59.5.1 Interrupt Handling and Event Linking

The DOC has a bit to enable or disable interrupts. When an interrupt source condition is satisfied while the interrupt is enabled, the interrupt request signal is issued to the CPU.
In contrast, an event link output signal is sent to other modules as an event signal via the ELC when an interrupt source is generated, regardless of the setting of the corresponding interrupt enable bit.

59.6 Usage Note

59.6.1 Module Stop Function Setting

Operation of the DOC can be enabled or disabled by setting the MSTPB6 bit in module stop control register B (MSTPCRB). The DOC is initially disabled after a reset. Register access is enabled by releasing the module stop state. For details, refer to section 11, Low Power Consumption.